Weather Income Protection for Delivery Workers

Problem statement:

Delivery workers depend heavily on daily earnings from completed deliveries. However, during extreme weather conditions such as heavy rainfall or high temperatures, delivery demand decreases significantly. This leads to reduced working hours and unstable income for workers.

This project proposes a micro-insurance system that provides financial protection to delivery workers by compensating them for temporary income loss caused by extreme weather conditions.

Target Users

The primary users of this solution are food delivery workers who operate through gig platforms such as food delivery services.

Characteristics of these workers include:

- They earn income based on the number of deliveries completed.
- Their earnings fluctuate depending on demand and working conditions.
- Extreme weather conditions often reduce order volumes.
- This results in unpredictable and unstable income.

Risk Triggers

Income loss is triggered when specific weather conditions exceed predefined thresholds.

The key risk triggers include:

Heavy Rain

- Rainfall greater than 50 mm

Extreme Heat

- Temperature greater than 40°C

When these conditions occur, delivery demand and worker activity decrease, leading to income loss.

Weekly Insurance Plans

Plan	Weekly Premium	Coverage
Basic	₹20	Covers up to 3 hours of lost income
Standard	₹40	Covers up to 6 hours of lost income
Premium	₹60	Covers up to 10 hours of lost income

These plans allow workers to choose coverage based on their risk tolerance and affordability.

Claim and Payout Process

The system uses automated triggers to ensure a simple and fast payout process.

Process flow:

1. A weather data API detects extreme weather conditions.
2. Delivery platform data shows reduced delivery activity.
3. The system estimates the worker's potential income loss.
4. The insurance platform automatically triggers a payout to the worker.

This removes the need for manual claims and ensures quick financial support.

AI Usage

Artificial Intelligence is used to improve the accuracy and reliability of the system.

AI can help with:

- Predicting weather-related risk levels.
- Estimating possible income loss for workers.
- Detecting suspicious or fraudulent claims.

Even simple machine learning models can be used to support these functions.

Architecture:

Worker Application

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Insurance Platform

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Weather Data + Delivery Platform Data

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AI Risk Analysis Model

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Automatic Payout System